

3.1 Waves + electromagnetic spectrum — 5054 Paper 2

Newest → oldest. 3.1 and EM-spectrum subquestions. Mark scheme clipped to those parts only — not the rest of the question.

2025

Oct/Nov 5054/22

Question 5 · included (a), (b)

EM map +  $v = f\lambda$

(a) Fig. 5.1 represents the electromagnetic spectrum divided into the seven main regions.

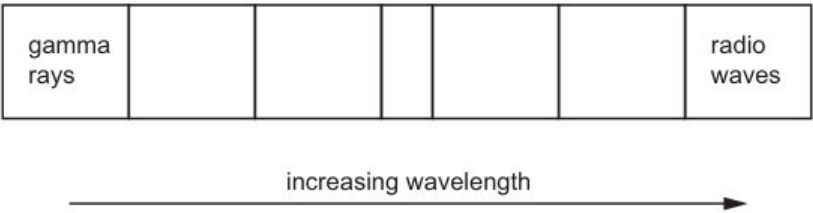


Fig. 5.1

The regions are arranged from left to right in order of increasing wavelength.

On Fig. 5.1:

- indicate the microwave region by marking it with the letter M
- indicate the ultraviolet region by marking it with the letters UV.

[1]

(b) The frequency of the microwaves used by some satellite television systems is 12 GHz.

Mark scheme (kept parts only) — 5054/22 Q5 (a), (b)

|         |                                                                                                                                          |    |
|---------|------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5(a)    | <div><div><div></div><div></div><div>UV</div><div></div><div></div><div>M</div><div></div></div><div>increasing wavelength →</div></div> | B1 |
| 5(b)(i) | $(\lambda =) \frac{c}{f} \text{ or } \frac{3.0 \times 10^8}{1.2 \times 10^N}$                                                            | C1 |

May/June 5054/22

Question 4 · included (a), (b), (c)

(a) Define 'power'.

.....  
 ..... [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power = .....W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

### Mark scheme (kept parts only) — 5054/22 Q4 (a), (b), (c)

|          |                                                                         |           |
|----------|-------------------------------------------------------------------------|-----------|
| 4(c)(i)  | (at least) three correctly curved wavefronts centered on gap            | <b>B1</b> |
|          | same wavelength as on left <b>and</b> some curving in correct direction | <b>B1</b> |
| 4(c)(ii) | diffraction                                                             | <b>B1</b> |

|         |                                                                                                                                                                                                                            |           |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)    | <i>motion of particles:</i> backwards and forwards / to and fro / closer and further apart                                                                                                                                 | <b>B1</b> |
|         | (oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork<br><b>or</b> mention of compressions and rarefactions | <b>B1</b> |
| 4(b)(i) | number of oscillations per second<br><b>or</b> number of wave(length)s (passing a point) per second                                                                                                                        | <b>B1</b> |

May/June 5054/21

Question 5 · included (a), (b), (c)

- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency = .....Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed .....

wavelength .....

[2]



- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]

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[Turn over]

**Mark scheme (kept parts only) — 5054/21 Q5 (a), (b), (c)**

|          |                                                                                                                        |           |
|----------|------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(b)(i)  | 0.75 (Hz)                                                                                                              | <b>B1</b> |
| 5(b)(ii) | <i>speed</i> : no change                                                                                               | <b>B1</b> |
|          | <i>wavelength</i> : decreases                                                                                          | <b>B1</b> |
| 5(c)     | spreading out / bending at an edge / obstacle<br><b>or</b><br>spreading out after passing through a (small) gap / hole | <b>B1</b> |

|          |                                                                                      |           |
|----------|--------------------------------------------------------------------------------------|-----------|
| 5(a)(i)  | C and R marked correctly                                                             | <b>B1</b> |
| 5(a)(ii) | compressions are where pressure is high / particles close together / density is high | <b>B1</b> |
| 5(b)(i)  | 0.75 (Hz)                                                                            | <b>B1</b> |
| 5(b)(ii) | <i>speed</i> : no change                                                             | <b>B1</b> |

**2024**

**Oct/Nov 5054/21**

**Question 5 · included (b), (c)**

*not vacuum*

(b) Describe how a longitudinal wave differs from a transverse wave.

.....

.....

..... [2]

(c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

### Mark scheme (kept parts only) — 5054/21 Q5 (b), (c)

|         |                                                                                              |           |
|---------|----------------------------------------------------------------------------------------------|-----------|
| 5(b)    | (in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction  | <b>B1</b> |
|         | in a transverse wave, vibration (of particles) is perpendicular to the propagation direction | <b>B1</b> |
| 5(c)(i) | $(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$                                              | <b>C1</b> |

### Question 4 · included (a), (b), (c), (d)

visible + IR

(a) (i) Calculate the initial gravitational potential energy of the ball.

gravitational potential energy = ..... J [2]

- (b) When the ball hits the ground, it stops moving.

The kinetic energy of the ball is transferred to internal energy.

The temperature of the ball increases.

- (i) The specific heat capacity of modelling clay is  $1400 \text{ J/(kg } ^\circ\text{C)}$ .

Calculate the maximum possible temperature increase of the ball. Show your working.

temperature increase = .....  $^\circ\text{C}$  [2]

- (ii) Suggest **one** reason why the actual temperature increase of the ball may be smaller than the value calculated in (b)(i).

.....  
.....  
..... [1]

[Total: 8]



- (c) The gas in the cylinder reaches a temperature of  $100^{\circ}\text{C}$ .

The piston is released and moves to the right. The temperature remains at  $100^{\circ}\text{C}$ .

The pressure of the gas now decreases.

- (i) Explain, in terms of the particles of the gas, why the pressure of the gas now decreases.

.....  
 .....  
 ..... [1]

- (ii) Eventually, the piston stops moving.

Explain why.

.....  
 .....  
 .....  
 ..... [2]

[Total: 10]

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[Turn over]

- (d) On Fig. 4.1, point P is shown immediately next to point R.

No light is observed at point P, but a detector at point P registers radiation reaching the detector.

State which region of the electromagnetic spectrum is detected at point P.

..... [1]

[Total: 8]

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### Mark scheme (kept parts only) — 5054/21 Q4 (a), (b), (c), (d)

|         |                                                                                                                   |           |
|---------|-------------------------------------------------------------------------------------------------------------------|-----------|
| 4(a)    | at least <b>one</b> ray from left-hand side showing correct refraction at both faces <b>and</b> no incorrect rays | <b>B1</b> |
|         | at least <b>two</b> correct rays (i.e. two rays within glass) <b>and</b> one ending at R and the other at V       | <b>B1</b> |
| 4(b)    | both (red and violet light) travel more slowly in glass than in air                                               | <b>B1</b> |
|         | red light travels more quickly (than violet light) in glass                                                       | <b>B1</b> |
| 4(c)(i) | any <b>four</b> (of the five) colours in order <b>or</b> five colours in any order                                | <b>C1</b> |
|         | red    orange    yellow    green    blue    indigo    violet                                                      | <b>B1</b> |

|      |                      |    |
|------|----------------------|----|
| 4(d) | infrared (radiation) | B1 |
|------|----------------------|----|

| Question | Answer | Marks |
|----------|--------|-------|
|----------|--------|-------|

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**Question 4 · included (a)(i), (a)(ii), (b), (c)**

*not ultrasound use*

**(b)** Sound waves are longitudinal waves. Water waves are transverse waves.

- (i)** Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

.....

.....

.....

.....

..... [3]

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- (i)** State what is meant by 'frequency'.

.....

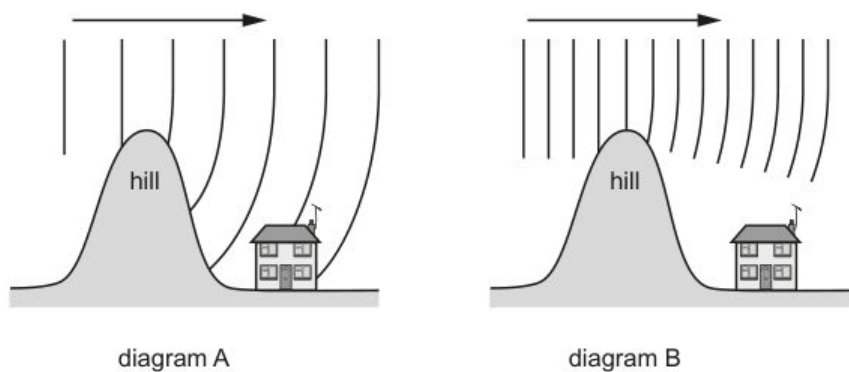
..... [1]

- (ii)** Calculate the speed of ultrasound in air.

Show your working.

speed = ..... m/s [2]

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.



**Fig. 4.1** (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

.....

.....

.....

..... [2]

[Total: 10]

**Mark scheme (kept parts only) — 5054/21 Q4 (a)(i), (a)(ii), (b), (c)**

*MS clip not separated — see official MS for the listed parts only.*

**2023**

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**Question 10 · included (b)**

*only  $v=f\lambda$*

- (b) The gamma radiation produced has a frequency of  $8.8 \times 10^{19}$  Hz.

The speed of electromagnetic radiation in a vacuum is  $3.0 \times 10^8$  m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = ..... m [2]

[Total: 9]

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### Mark scheme (kept parts only) — 5054/22 Q10 (b)

|       |                                                          |           |
|-------|----------------------------------------------------------|-----------|
| 10(b) | $(\lambda =) v/f = 3.0 \times 10^8 / 8.8 \times 10^{19}$ | <b>C1</b> |
|       | $3.4 \times 10^{-12}$ (m)                                | <b>A1</b> |

**Question 5 · included (a), (b)***EM + trans/long*

- (b)**
- Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

- (a)**
- The name of one of these three regions is the ultraviolet region.

- (i)**
- State the names of all three regions in order of increasing frequency.

smallest frequency .....

**Mark scheme (kept parts only) — 5054/21 Q5 (a), (b)**

|          |                                                                                                             |           |
|----------|-------------------------------------------------------------------------------------------------------------|-----------|
| 4(b)(i)  | speed: decreases<br>frequency: stays the same<br>wavelength: decreases                                      | <b>B2</b> |
| 5(a)(i)  | infrared (radiation) <b>and</b> visible (light) <b>and</b> ultraviolet (radiation)                          | <b>B1</b> |
|          | infrared (radiation) <b>and</b> visible (light) <b>and</b> ultraviolet (radiation) <b>and</b> in this order | <b>B1</b> |
| 5(a)(ii) | damaging effect: (skin) cancer / cataracts                                                                  | <b>B1</b> |
|          | property: it / ultraviolet radiation is ionising                                                            | <b>B1</b> |

**Question 4 · included (a), (b)(i)***not n or rays*

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed .....

frequency .....

wavelength .....

[2]

- (a) Calculate the wavelength in air of this red light.

wavelength = ..... m [3]

### Mark scheme (kept parts only) — 5054/21 Q4 (a), (b)(i)

|         |                                                                        |    |
|---------|------------------------------------------------------------------------|----|
| 4(b)(i) | speed: decreases<br>frequency: stays the same<br>wavelength: decreases | B2 |
| 4(a)    | $(\lambda =) v / f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$          | C1 |
|         | $6.4 \times 10^{-7}$ (m) or $3 / 4.7 \times 10^6$                      | C1 |
|         | $6.4 \times 10^{-7}$ (m)                                               | A1 |

### May/June 5054/22

#### Question 7 · included (a)

UV / shorter  $\lambda$

- 7 (a) Ultraviolet radiation is one component of the electromagnetic spectrum.

- (i) State the name of **two** components of the electromagnetic spectrum that have a smaller wavelength than ultraviolet radiation.

1 .....

### Mark scheme (kept parts only) — 5054/22 Q7 (a)

MS clip not separated — see official MS for the listed parts only.

#### Question 5 · included (a), (b)

- (a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

- (b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

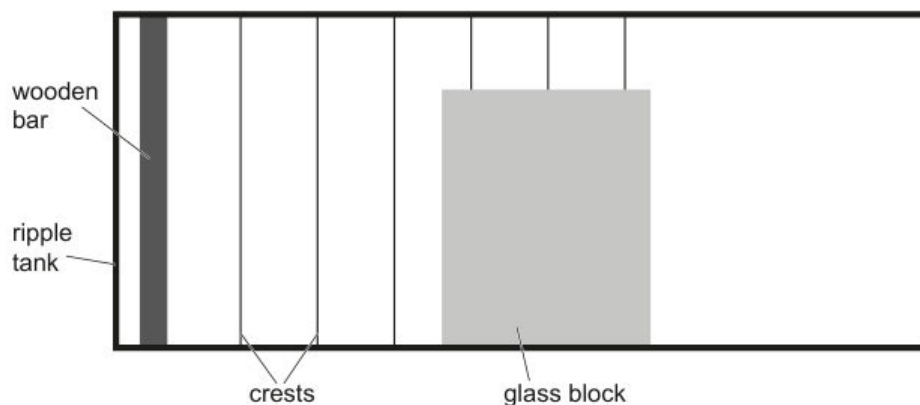


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

### Mark scheme (kept parts only) — 5054/22 Q5 (a), (b)

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May/June 5054/21

### Question 7 · included (a), (b), (c)

- (a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

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- (b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = ..... kg [3]

- (c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

distance = ..... m [2]

**Mark scheme (kept parts only) — 5054/21 Q7 (a), (b), (c)**

*MS clip not separated — see official MS for the listed parts only.*

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**2022**

**Oct/Nov 5054/22**

**Question 4 · included (b), (c)**

*not vacuum; not hearing*

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength  $g$  is 10 N/kg.

(c) Describe what is meant by 'a rarefaction'.

.....  
 ..... [1]

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### Mark scheme (kept parts only) — 5054/22 Q4 (b), (c)

|         |                                                                                                                                                                                            |           |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(b)(i) | radiation <b>or</b> infrared mentioned                                                                                                                                                     | <b>B1</b> |
|         | radiation <b>and</b> infrared (travel to the man)                                                                                                                                          | <b>B1</b> |
|         | energy / infrared / radiation <u>absorbed</u> by the man / pullover <b>or</b> travels in straight line <b>or</b> travels at the speed of light <b>or</b> increases (man's) internal energy | <b>B1</b> |
| 4(c)    | (point where) molecules are further apart (than average) <b>or</b><br>(point where) pressure / density is lower (than average)                                                             | <b>B1</b> |

Oct/Nov 5054/21

### Question 4 · included (a), (b), (c)

*X-rays as EM*

(a) Calculate the change in gravitational potential energy (g.p.e.) of the car as it travels from A to B.

change in g.p.e. = ..... [2]

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- (b) Calculate the speed of the car when it reaches B.

speed = ..... [3]

- (c) After releasing it, the child expects the car to follow the track along the route ABCDEFG. In fact, the model car does not reach F.

### Mark scheme (kept parts only) — 5054/21 Q4 (a), (b), (c)

|          |                                                                                                                                                                                      |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 4(c)(i)  | X-rays directed towards body part with (suspected) broken bone                                                                                                                       | <b>B1</b> |
|          | X-rays pass through flesh but not (to same extent) through bones                                                                                                                     | <b>B1</b> |
|          | image produced by digital / electronic / CCD / CMOS detector                                                                                                                         | <b>B1</b> |
| 4(c)(ii) | X-rays (may) cause cancer / cell mutation / birth defect / miscarriage <b>or</b><br>X-rays show the bone structure rather than foetal development <b>or</b> image would not be clear | <b>B1</b> |

|         |                                                                          |           |
|---------|--------------------------------------------------------------------------|-----------|
| 4(a)    | electromagnetic (waves) <b>or</b> components of electromagnetic spectrum | <b>B1</b> |
| 4(b)(i) | number of oscillations / cycles / wavelengths                            | <b>B1</b> |
|         | per unit time                                                            | <b>B1</b> |

May/June 5054/22

### Question 8 · included (a), (b), (c)

*not satellite*

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....  
 ..... [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed .....  
 wavelength ..... [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.

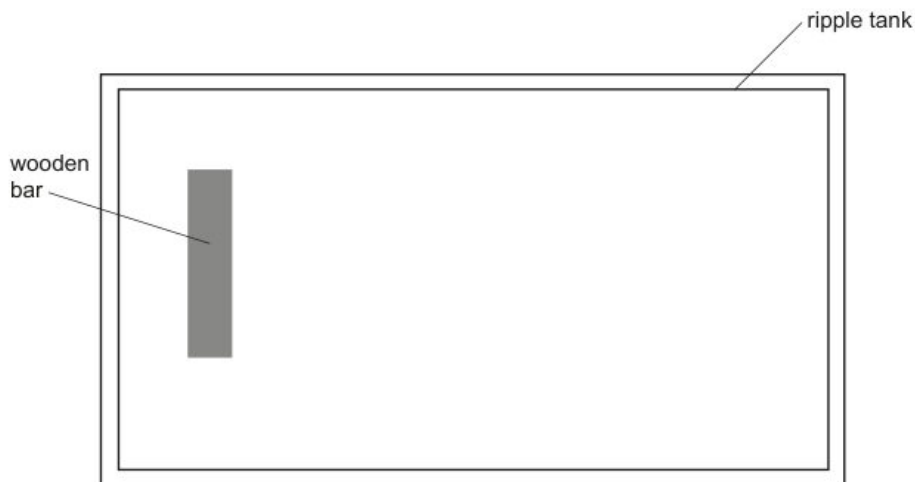


Fig. 8.3

[3]

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

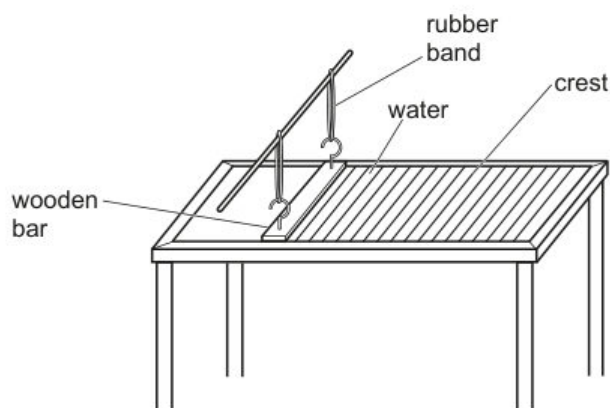


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.



Fig. 8.3

[3]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....  
..... [1]

**Mark scheme (kept parts only) — 5054/22 Q8 (a), (b), (c)**

|  |           |  |    |
|--|-----------|--|----|
|  | 0.00007 s |  | 21 |
|--|-----------|--|----|

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**2021**

**Oct/Nov 5054/22**

**Question 3 · included (a)**

*not TIR*

- (a) Light travels more slowly in the block than in air.

- (i) Explain how Fig. 3.1 shows this.

.....

**Mark scheme (kept parts only) — 5054/22 Q3 (a)**

*MS clip not separated — see official MS for the listed parts only.*

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**Oct/Nov 5054/21**

**Question 8 · included (b)(iii), (b)(v)**

*only long/trans and  $v=f\lambda$*

- (b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

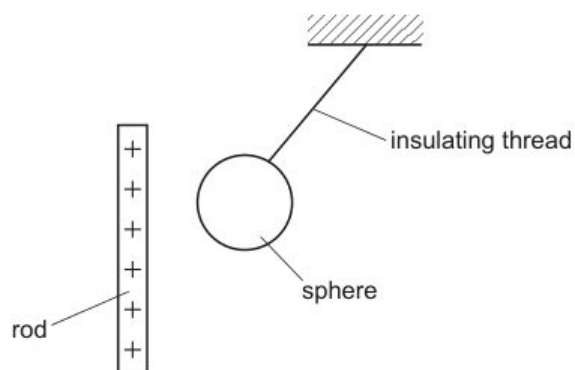


Fig. 5.1

- (i) On Fig. 5.1, draw the distribution of charge on the sphere. [2]
- (ii) Explain why the sphere is pulled towards the rod.

.....

.....

.....

..... [2]

[Total: 6]

**Mark scheme (kept parts only) — 5054/21 Q8 (b)(iii), (b)(v)**

*MS clip not separated — see official MS for the listed parts only.*

**May/June 5054/22**

**Question 10 · included (a), (b)**

*water + visible*

- 10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

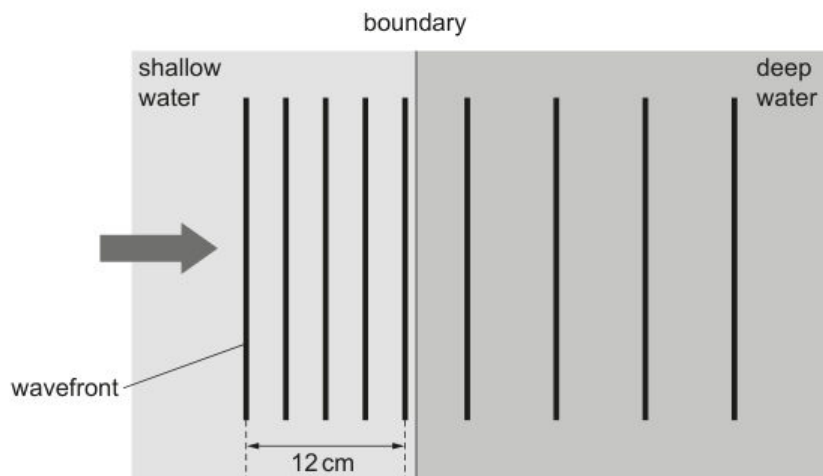


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (b) White light is made up of different colours.
- (i) State the name of four of the colours in the visible spectrum and place them in order from the smallest wavelength to the largest wavelength.

smallest wavelength .....

### Mark scheme (kept parts only) — 5054/22 Q10 (a), (b)

MS clip not separated — see official MS for the listed parts only.

May/June 5054/21

### Question 4 • included (a)

- 4 (a) Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.



Fig. 4.1

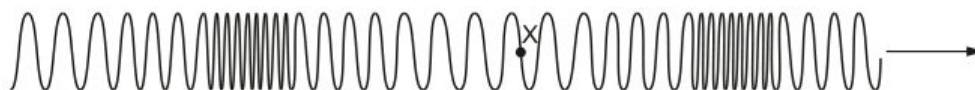


Fig. 4.2

### Mark scheme (kept parts only) — 5054/21 Q4 (a)

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Oct/Nov 5054/21

### Question 4 • included (a)

not TIR

(a) Light travels more slowly in glass than in air.

(i) State what happens to the wavelength of the light in the ray as it enters the glass.

..... [1]

**Mark scheme (kept parts only) — 5054/21 Q4 (a)**

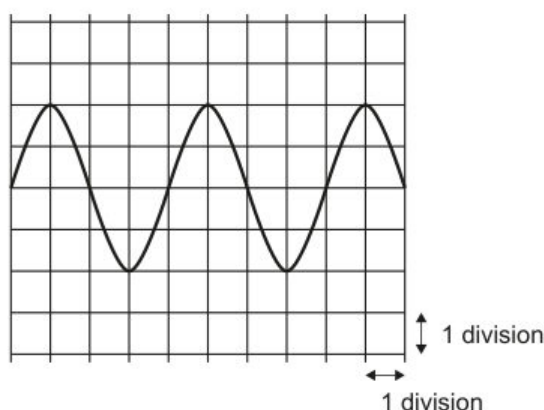
*MS clip not separated — see official MS for the listed parts only.*

**May/June 5054/22**

**Question 10 · included (b), (c)(i), (c)(ii)**

*not thermionic/pitch/diode*

(b) Fig. 10.1 is the trace on the screen of the oscilloscope.



**Fig. 10.1**

The settings on the oscilloscope are 10 ms/division for the x-axis and 3.0 V/division for the y-axis.

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(c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

(i) Sound is a type of wave.

State which type.

..... [1]

(ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

.....  
.....  
..... [2]

**Mark scheme (kept parts only) — 5054/22 Q10 (b), (c)(i), (c)(ii)**

*MS clip not separated — see official MS for the listed parts only.*

### Question 5 · included (a)

EM uses

- (a) Draw **one** line from each component of the spectrum to another suitable use for that component.

| component   | use                             |
|-------------|---------------------------------|
|             | sunbeds                         |
| microwaves  | television controller           |
|             | satellite television            |
| ultraviolet | airport security check of cases |

[2]

### Mark scheme (kept parts only) — 5054/22 Q5 (a)

MS clip not separated — see official MS for the listed parts only.

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### Question 9 · included (a)(ii)

only X-ray  $v=f\lambda$

- (ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....  
..... [2]

### Mark scheme (kept parts only) — 5054/21 Q9 (a)(ii)

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2019

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### Question 5 · included (a), (b)(i)

not cleaning

- (i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = ..... [2]

(a) Describe what is meant by *longitudinal*.

.....

.....

..... [2]

**Mark scheme (kept parts only) — 5054/22 Q5 (a), (b)(i)**

*MS clip not separated — see official MS for the listed parts only.*

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**Oct/Nov 5054/21**

**Question 10 · included (a), (b), (c), (d)**

*EM + wavefronts*

- (iv) Determine the magnification of the image.

magnification = ..... [1]

- (v) State **two** ways in which a virtual image differs from a real image.

1. ....  
 .....  
 2. ....  
 ..... [2]

- (vi) State one use for a diverging lens.

.....  
 ..... [1]

- (d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

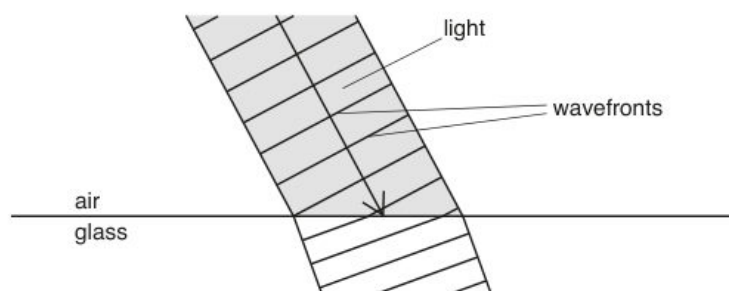


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....  
 .....  
 ..... [3]

[Total: 15]

(a) State the speed of light in a vacuum.

..... [1]

(b) State the colours of the visible spectrum.

..... [1]

(c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses),  $F_1$  and  $F_2$ , of the lens.

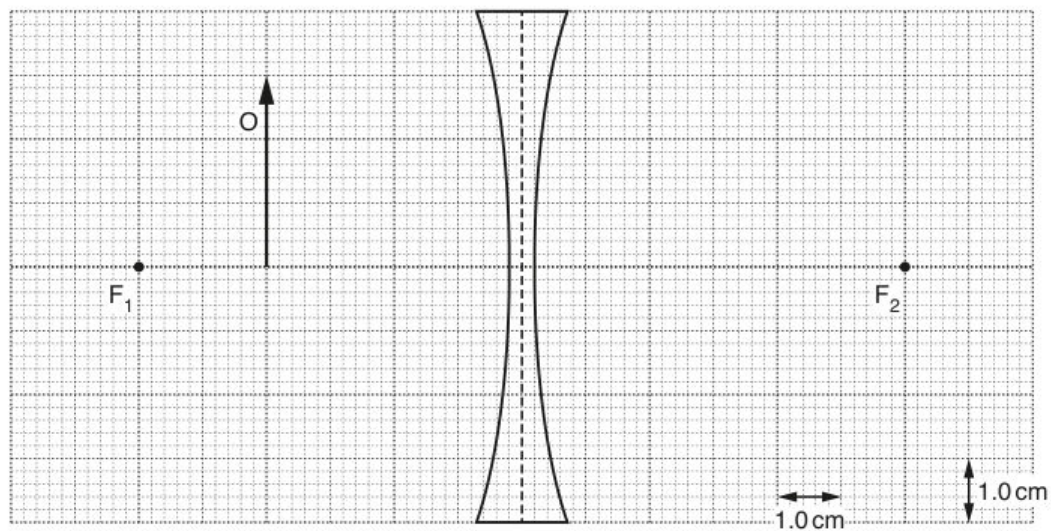


Fig. 10.1

- (d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

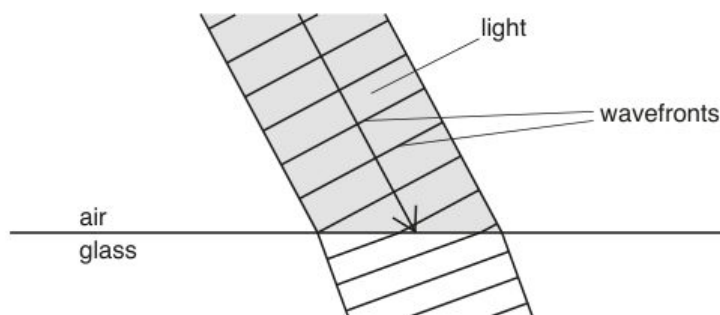


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

.....

..... [3]

[Total: 15]

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### Mark scheme (kept parts only) — 5054/21 Q10 (a), (b), (c), (d)

MS clip not separated — see official MS for the listed parts only.

#### Question 4 · included (a), (b)

EM labels

- (a) State the name of the component labelled:

- A .....
- B .....
- C ..... [2]

- (b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency .....

the smallest wavelength ..... [1]

### Mark scheme (kept parts only) — 5054/21 Q4 (a), (b)

MS clip not separated — see official MS for the listed parts only.

May/June 5054/22

#### Question 4 · included (a), (b), (c)

- (a) Define the term *pressure*.

.....  
 ..... [1]

- (b) The density of mercury is  $1.4 \times 10^4 \text{ kg/m}^3$ . The gravitational field strength  $g$  is  $10 \text{ N/kg}$ .

The pressure of the atmosphere is  $1.0 \times 10^5 \text{ Pa}$ .

Calculate the pressure of the gas in the container.

pressure = ..... [3]

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- (c) In Fig. 2.1, the mercury level on the left-hand side of the manometer is lower than on the right-hand side.

The gas inside the container is heated. This causes the mercury levels on both sides to become the same.

### Mark scheme (kept parts only) — 5054/22 Q4 (a), (b), (c)

MS clip not separated — see official MS for the listed parts only.

May/June 5054/21

### Question 10 · included (a)

not speed-of-sound exp

- (a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = ..... [2]

### Mark scheme (kept parts only) — 5054/21 Q10 (a)

MS clip not separated — see official MS for the listed parts only.

2018

Oct/Nov 5054/22

### Question 9 · included (b), (c)

not sound vs ultra

- (b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.
- (i) Explain what is meant by a wave and describe how a longitudinal wave differs from a transverse wave.
- .....
- (c) A sound wave travelling in a liquid, passes into air.
- (i) State what happens to the speed of the sound wave as it enters the air.
- ..... [1]

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5054/22/O/N/18

### Mark scheme (kept parts only) — 5054/22 Q9 (b), (c)

*MS clip not separated — see official MS for the listed parts only.*

Oct/Nov 5054/21

### Question 9 · included (a), (b)

*not TIR*

- (a) State how a longitudinal wave differs from a transverse wave.
- .....
- ..... [1]
- (b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

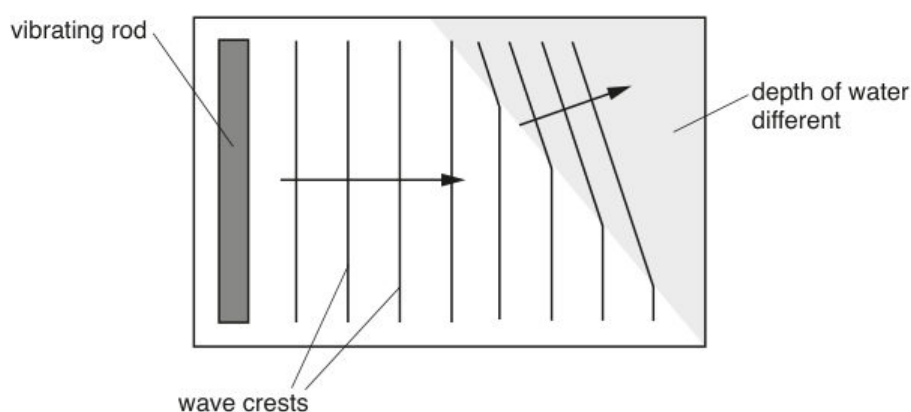


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

### Mark scheme (kept parts only) — 5054/21 Q9 (a), (b)

*MS clip not separated — see official MS for the listed parts only.*

May/June 5054/22

### Question 5 · included (b)

- (b) The object has a constant speed for some of the time.
- (i) State and explain how a distance-time graph shows that speed is constant.

.....

### Mark scheme (kept parts only) — 5054/22 Q5 (b)

MS clip not separated — see official MS for the listed parts only.

May/June 5054/21

#### Question 11 · included (a), (b)

- (b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16 °C

temperature of water after heating = 37 °C

volume of water collected in 1.0 minute =  $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = 4200 J / (kg °C)

density of water = 1000 kg / m<sup>3</sup>

- (i) Calculate the mass of water leaving the shower in 1.0 s.

mass = .....[2]

- 10 (a) (i) Describe two differences between boiling and evaporation.

1. ....

.....

.....

2. ....

.....

.....

[4]

## Mark scheme (kept parts only) — 5054/21 Q11 (a), (b)

MS clip not separated — see official MS for the listed parts only.

### Question 9 · included (a), (c)

EM T/F + speed

(c) The Earth is  $1.5 \times 10^8$  km from the Sun.

(i) State the speed of light in a vacuum.

.....[1]

(a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

|                                                                                 | true                     | false                    |
|---------------------------------------------------------------------------------|--------------------------|--------------------------|
| Gamma rays are used to kill cancerous cells but can also cause cancer.          | <input type="checkbox"/> | <input type="checkbox"/> |
| Infra-red is used in sun-beds.                                                  | <input type="checkbox"/> | <input type="checkbox"/> |
| Radio waves have the highest frequency in the electromagnetic spectrum.         | <input type="checkbox"/> | <input type="checkbox"/> |
| The higher the frequency of the radiation the smaller is the wavelength in air. | <input type="checkbox"/> | <input type="checkbox"/> |

[2]

## Mark scheme (kept parts only) — 5054/21 Q9 (a), (c)

MS clip not separated — see official MS for the listed parts only.

2017

Oct/Nov 5054/22

### Question 6 · included (a), (b), (c)

(a) State what is meant by the term *echo*.

.....  
.....[1]

(b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength = .....[2]

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5054/22/O/N/17

**Mark scheme (kept parts only) — 5054/22 Q6 (a), (b), (c)**

*MS clip not separated — see official MS for the listed parts only.*

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**Oct/Nov 5054/21**

**Question 10 · included (a), (b), (c)**

*EM +  $v=f\lambda$*

- (a) State the name of the component of the electromagnetic spectrum that is found in region

(i) P, .....

(ii) Q, .....

- (b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]

- (c) The speed of electromagnetic radiation in a vacuum is  $3.0 \times 10^8 \text{ m/s}$ .

A television remote controller uses infra-red radiation of wavelength  $9.4 \times 10^{-7} \text{ m}$ .

- (i) Calculate the frequency of this radiation.

frequency = ..... [3]

- (ii) Explain how infra-red radiation is used in television remote controllers.

.....  
.....  
.....  
..... [2]

**Mark scheme (kept parts only) — 5054/21 Q10 (a), (b), (c)**

*MS clip not separated — see official MS for the listed parts only.*

**May/June 5054/22**

**Question 6 · included (a), (b)(i)**

*not hearing*

- (i) Calculate the wavelength of the sound.

wavelength = ..... [2]

- 6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

**Mark scheme (kept parts only) — 5054/22 Q6 (a), (b)(i)**

*MS clip not separated — see official MS for the listed parts only.*

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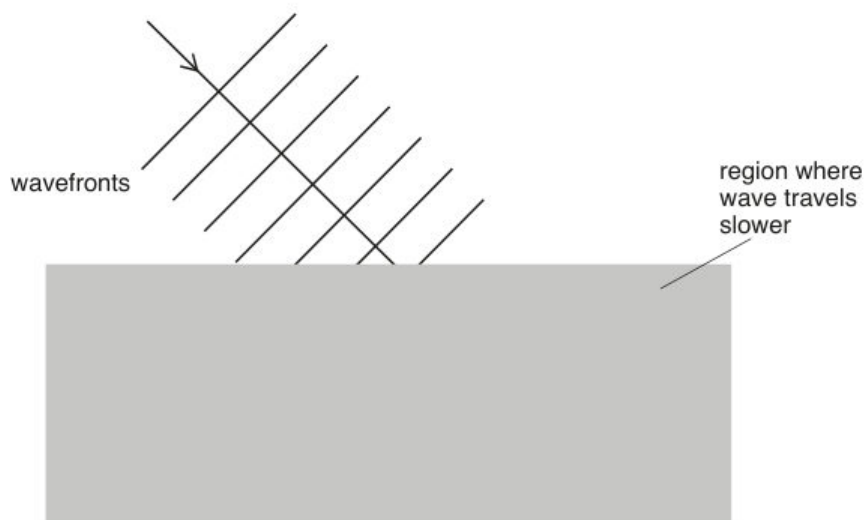
**May/June 5054/21**

**Question 10 • included (b), (c)(i), (c)(iii)**

*not mirror; not hearing*

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.



**Fig. 10.2**

- (i) State what is meant by *wavefront*.

.....

.....[2]

- (i) Calculate the speed of sound in air.

speed = .....[2]

- (ii) 1. State the range of frequencies that can be heard by a healthy human ear.

.....[1]

2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength = .....[1]

- (iii) Describe a simple experiment to show that sound waves obey the law of reflection.  
You may draw a diagram if you wish.

.....  
.....  
.....  
.....[3]

**Mark scheme (kept parts only) — 5054/21 Q10 (b), (c)(i), (c)(iii)**

---

**2016**

**Oct/Nov 5054/22**

**Question 10 · included (a)**

*only  $v=f\lambda$*

- (a) Calculate the speed of the skier after 5.0 s.

speed = .....[2]

### Mark scheme (kept parts only) — 5054/22 Q10 (a)

- |            |                                                                                                      |          |     |
|------------|------------------------------------------------------------------------------------------------------|----------|-----|
| 10 (a) (i) | $3.0 \times 10^8 \text{ m/s}$                                                                        | B1       |     |
| (ii)       | $(\lambda = )c/f \text{ or } 3.0 \times 10^8 / 4.3 \times 10^{14}$<br>$7.0 \times 10^{-7} \text{ m}$ | C1<br>A1 | [3] |

Oct/Nov 5054/21

### Question 5 · included (a), (b)

*not vacuum*

- (a) State what is meant by the *frequency* of a sound.

.....  
 .....  
 .....[1]

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength = .....[2]

### Mark scheme (kept parts only) — 5054/21 Q5 (a), (b)

- |         |                                                                                                                 |          |
|---------|-----------------------------------------------------------------------------------------------------------------|----------|
| 5 (a)   | number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time | B1       |
| (b) (i) | $(\lambda = )c/f \text{ or } 330 / 2200$<br>$0.15 \text{ m}$                                                    | C1<br>A1 |

May/June 5054/22

### Question 10 · included (b)

*CRO A and T*

- (b) A microphone is connected to a c.r.o. to display a sound wave.

Fig. 10.2 shows the trace on the c.r.o.

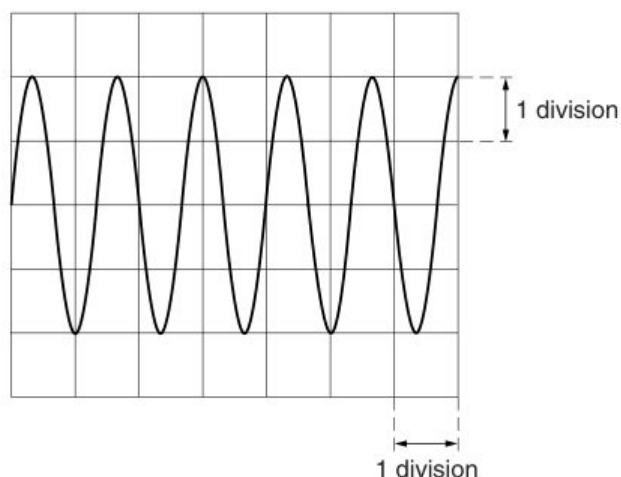


Fig. 10.2

The settings on the c.r.o. are: Y-gain 0.5V/division; time base 2.0ms/division.

### Mark scheme (kept parts only) — 5054/22 Q10 (b)

- |         |   |                                                                       |          |
|---------|---|-----------------------------------------------------------------------|----------|
| (b) (i) | 1 | 1(.0)V                                                                | B1       |
|         | 2 | one wave 1.3–1.4 squares <b>or</b> 3 waves in 4 squares<br>2.6–2.8 ms | C1<br>A1 |

### Question 6 · included (a), (b)

EM colours

- (a) State which colour in Fig. 6.1 has
- the largest wavelength, ..... [1]
  - the highest frequency. .... [1]

- (b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component .....

description of use .....

.....

.....

2. component .....

description of use .....

.....

.....

[4]

**Mark scheme (kept parts only) — 5054/22 Q6 (a), (b)**

- |                                            |                                                                     |    |
|--------------------------------------------|---------------------------------------------------------------------|----|
| 4                                          | <p>(a) steam<br/>or (water) vapour<br/>or water in gaseous form</p> | B1 |
| <p>(b) (E =) mL numerical or algebraic</p> |                                                                     | C1 |

**May/June 5054/21**

**Question 7 · included (c)**

only  $v=f\lambda$

- (c) The loudspeaker produces sound of frequency 500Hz. The speed of sound is 320m/s. Calculate the wavelength of the sound produced.

wavelength = .....[2]

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### Mark scheme (kept parts only) — 5054/21 Q7 (c)

MS clip not separated — see official MS for the listed parts only.

### Question 5 · included (a), (b)

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

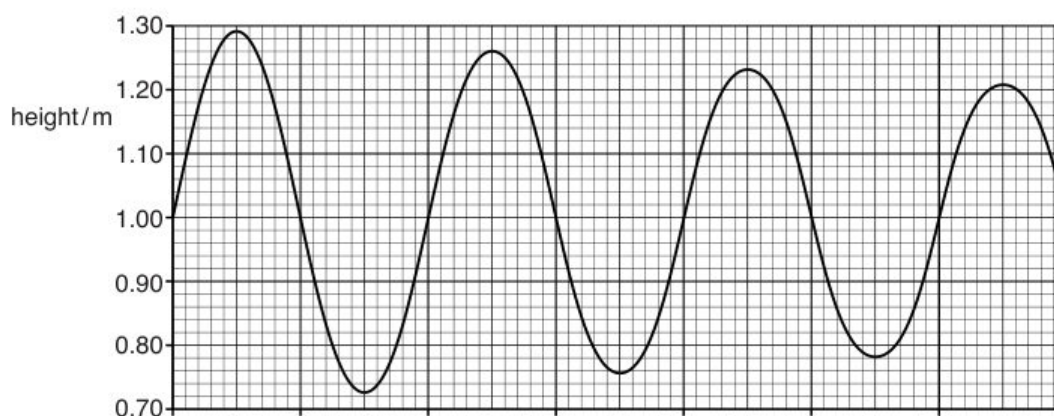
Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

| wave  | direction of vibration                                   | type of wave |
|-------|----------------------------------------------------------|--------------|
| light | perpendicular to the direction of transfer of the energy | transverse   |
| sound |                                                          |              |
| water |                                                          |              |

Fig. 5.1

[2]

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.



### Mark scheme (kept parts only) — 5054/21 Q5 (a), (b)

- |     |      |                                                                                                                                                                                                                                                            |    |
|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 5   | (a)  | <i>sound</i> : along or parallel (to transfer of energy or wave) <b>and</b> longitudinal                                                                                                                                                                   | B1 |
|     |      | <i>water</i> : perpendicular <b>and</b> transverse                                                                                                                                                                                                         | B1 |
| (b) | (i)  | 0.29 – 0.28 m                                                                                                                                                                                                                                              | B1 |
|     | (ii) | <u>time/period</u> for one wave(length)/cycle constant<br>or each oscillation/cycle takes one second                                                                                                                                                       | B1 |
| (b) | (i)  | 1 difference between smallest and largest temperature<br>or from 0 to 100 °C                                                                                                                                                                               | B1 |
|     | (i)  | 2 small/moderate distance between (thermometer) marks<br>or for a given temperature change there is a small expansion of liquid /distance<br>(along scale)/change in thermometric property<br>or cannot measure small temperature <u>difference/change</u> | B1 |

2015

Oct/Nov 5054/21

**Question 7 · included (a), (b)**

*not ultrasound use*

- (a) Describe one difference between a compression and a rarefaction.

.....

.....

..... [1]

- (b) In a particular medium, the speed of both audible sound and ultrasound waves is  $v$ .

**Mark scheme (kept parts only) — 5054/21 Q7 (a), (b)**

- |   |     |                                                                                                                         |          |
|---|-----|-------------------------------------------------------------------------------------------------------------------------|----------|
| 5 | (a) | <b>temperature</b> at which a liquid becomes a gas                                                                      | B1       |
|   | (b) | (i) molecules close together / touching <b>or</b> closer than in gas<br>randomly arranged <b>or</b> irregular structure | B1<br>B1 |

May/June 5054/22

**Question 9 · included (a), (b), (c), (d)**

- (a) Explain what is meant by the *frequency* of a wave.

.....

.....

..... [2]

- (b) (i) Determine the wavelength of the wave in deep water.

..... [1]

- (c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

- (i) State and explain how this affects the wavelength of the wave.

.....  
 .....  
 ..... [2]

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5054/22/M/J/15

- (d) Sound is also a wave.

- (i) Describe one difference between a sound wave and a water wave.

.....

### Mark scheme (kept parts only) — 5054/22 Q9 (a), (b), (c), (d)

- |       |                                                                                                                                                                            |          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 7 (a) | to provide a complete circuit (with live)<br>or to pass current back to mains<br>or provide a return path for the current                                                  | B1       |
| (b)   | current / charge / electrons flow to earth / earth wire / ground (when live touches case)<br>fuse melts / blows <b>and</b> disconnects circuit / cuts live / stops current | B1<br>B1 |
| (c)   | doubly insulated<br>or <b>case / body</b> made of plastic / insulator / not made of metal<br>or user cannot touch metal                                                    | B1       |
| (d)   | (circuit breaker)<br>• turns off / acts fast(er)                                                                                                                           | B1       |

May/June 5054/21

### Question 4 · included (a), (b), (c)

- (a) Describe the movement of the particles of the string.

.....  
 .....  
 .....  
 ..... [2]

- (b) Determine the wavelength of the wave.

wavelength = ..... [1]

- (c) The speed of the wave is 2.0 m/s.

**Mark scheme (kept parts only) — 5054/21 Q4 (a), (b), (c)**

|   |         |                                                                    |    |
|---|---------|--------------------------------------------------------------------|----|
|   | (b) (i) | straight line from origin to 200 m at 40 s                         | B1 |
|   |         | any line straight or curved from (40,200) to (60,500)              | B1 |
|   | (ii)    | $s = d/t$ or $500/60$                                              | C1 |
|   |         | 8.3 m/s                                                            | A1 |
|   | (c)     | molecules hit sides/piston                                         | B1 |
|   |         | more molecules hit per second/hit more frequently                  | B1 |
|   |         | molecular impacts create large(r) <b>force</b> (upwards on piston) | B1 |
| 1 | (a) (i) | 60 m                                                               | B1 |
|   |         | (ii) 12 s                                                          | B1 |
|   | (b) (i) | straight line from origin to 200 m at 40 s                         | B1 |
|   |         | any line straight or curved from (40,200) to (60,500)              | B1 |

---